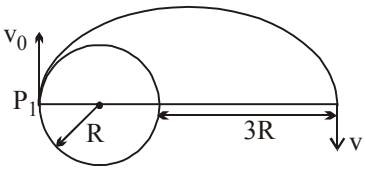
SOLUTION
SECTION-I

Single Correct Answer Type **8 Q. [3 M (–1)]**

1. Ans. (A)

Sol.



Let speed at max height is v

Since, it is at max height, v is perpendicular to r

$\Rightarrow$ By conservation of angular momentum

$mv_0R = mv(4R)$

$$v = \frac{v_0}{4}$$

$\Rightarrow$ By conservation of mechanical energy

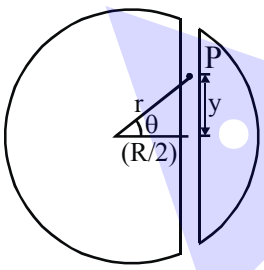
$$\frac{1}{2}mv_0^2 - \frac{GMm}{R} = \frac{1}{2}m\left(\frac{v_0}{4}\right)^2 - \frac{GMm}{4R}$$

$$v_0 = \sqrt{\frac{8}{5}\frac{GM}{R}}$$

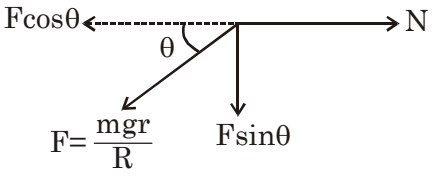
$$g = \frac{GM}{R^2}, v_0 = \sqrt{\frac{8}{5}gR}$$

2 Ans. (D)

Sol.



Gravity at point $P = \left(\frac{gr}{R}\right)$



$F\cos\theta - N = 0$

$F\sin\theta = ma$

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$$\frac{mgr}{R}\left(\frac{y}{r}\right)=ma$$

$$a=\left(\frac{g}{R}\right)y=\frac{g}{R}\sqrt{r^2-\left(\frac{R}{2}\right)^2}$$

$a \to f(r)$

3. Ans. (C)

Sol. $v_e=\sqrt{2}\sqrt{\frac{GM}{r_0}}$

$$\therefore \Delta v=v_e-v_0=\left(\sqrt{2}-1\right)\sqrt{\frac{GM}{r_0}}$$

4. Ans. (D)

Sol. Total mechanical energy = 0

$$-2\frac{GM}{a}m+\frac{1}{2}mv_e^2=0$$

$$v_e=2\sqrt{\frac{GM}{a}}$$

5. Ans. (B)

Sol. Using conservation of angular momentum about O.

$mV_Pr_P=mV_Ar_A=mV_0r_0\cos\theta$

$$V_Ar_A=V_Pr_P=\frac{3v_0r_0}{5}$$

using conservation of energy

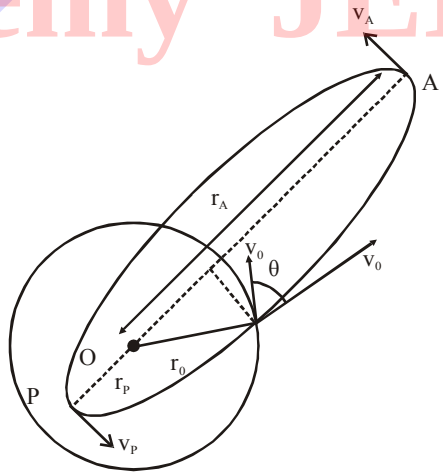
$$\frac{1}{2}mV_A^2+\frac{-GMm}{r_A}=\frac{-GMm}{r_0}+\frac{1}{2}mV_0^2$$

$$\Rightarrow \frac{9V_0^2r_0^2}{50r_A^2}-\frac{V_0^2r_0}{r_A}+\frac{V_0^2}{2}=0 \quad \left[\text{Let}\frac{r_0}{r_A}=x\right]$$

$$\Rightarrow 9x^2-50x+25=0 \quad \Rightarrow \quad x=5 \text{ or } (5/9)$$

$$\Rightarrow r_A=\frac{9r_0}{5}\ ;r_P=\frac{r_0}{5}$$

$$\Rightarrow \frac{V_P}{V_A}=\frac{r_A}{r_P}=9$$



6. Ans. (D)

Sol. Speed $=\sqrt{\frac{GM}{r}}$, Time period $=\left(\frac{2\pi}{\sqrt{GM}}\right)r^{3/2}$, Total energy $=-\frac{GMm}{2r}$

7. Ans. (A)

Sol. Areal velocity $=\frac{dA}{dt}=\frac{v_0r}{2}$

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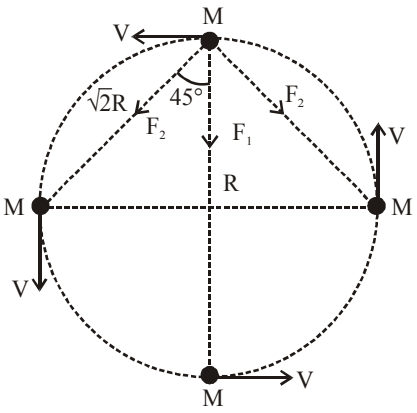
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& $V_0 = \sqrt{\frac{GM}{r}}$, V_0 = orbital speed

r = radius of orbit

8. Ans. (B)

Sol.



Net force on one particle

$F_{net} = F_1 + 2F_2 \cos 45° =$ Centripetal force

$$\Rightarrow \frac{GM^2}{(2R)^2} + \left[\frac{2GM^2}{(\sqrt{2}R)^2}\cos 45°\right] = \frac{MV^2}{R}$$

$$V = \frac{1}{2}\sqrt{\frac{GM}{R}(1+2\sqrt{2})}$$

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Multiple Correct Answer Type 7 Q. [4 M (–1)]

9. Ans. (A, B, C)

Sol. For (A) : By COME total mechanical energy of the two objects

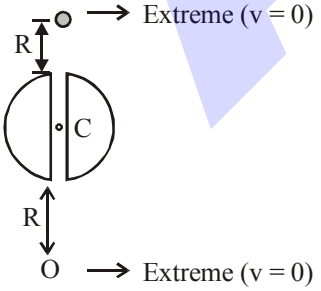
For (B) : By using reduced mass concept

$$\frac{1}{2}\mu v_{rel}^2 - \frac{G(4m)(m)}{r} = 0 \text{ where } \mu = \frac{(m)(4m)}{m+4m} = \frac{4}{5}m \Rightarrow v_{rel} = \sqrt{\frac{10Gm}{r}}$$

For (C) total KE = –PE = $\frac{G(m)(4m)}{r} = \frac{4Gm^2}{r}$

10. Ans. (A,C, D)

Sol.



Let r is distance from center,

If r > R $\quad F = \frac{GMm}{r^2}$ (Not SHM)

r < R $\quad F = \frac{GMmr}{R^3}$ (SHM)

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$\Rightarrow$ But motion is periodic to find velocity at center,
Apply COM E,

$$\frac{1}{2}m\left(0^2\right)-\frac{GMm}{2R}=\frac{1}{2}mv_C^2-\frac{3}{2}\frac{GMm}{R}$$

$$v_C=\sqrt{\frac{2GM}{R}}$$

11. Ans. (A,C)

Sol. $V_{CM}=0$, $Mv_1+2Mv_2=0$

$\frac{1}{2}Mv_1^2+\frac{1}{2}2Mv_2^2=\frac{GM(2M)}{3R}-0\Rightarrow$ relative velocity at t = T is $\frac{1}{3}\sqrt{\frac{2GM}{R}}$

12. Ans. (A,B,C,D)

13. Ans. (A, D)

Sol. ω in both is same
so $T_1=T_2$
and by $v_{max}=A\omega$ $\quad V_1>V_2$

14. Ans. (A,B,C,D)

Sol. $V=\sqrt{\frac{GM}{r}}$

$$\frac{V_A}{V_B}=\sqrt{\frac{r_B}{r_A}}=2$$

$$E=\frac{-GMm}{2r},\qquad \frac{E_A}{E_B}=\frac{m_A}{m_B}\times\frac{r_B}{r_A}=12$$

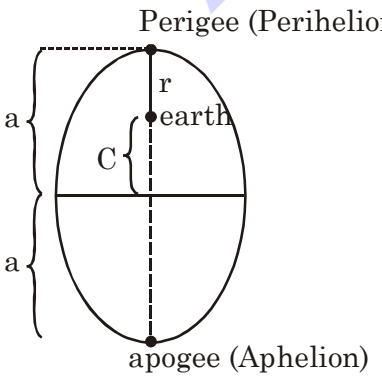
$$u_A=\sqrt{\frac{2GM}{r}},\qquad \frac{u_A}{u_B}=2$$

$$T=\frac{2\pi}{\sqrt{GM}}(r)^{3/2}$$

$$\frac{T_A}{T_B}=\left(\frac{r_A}{r_B}\right)^{3/2}=\frac{1}{8}$$

15. Ans. (B,C)

Sol. Initially parabolic path means escape velocity & hence (TE = 0)



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$$r_{min} = (a - c) = \frac{a}{2}$$

$$r_{max} = (a + c) = \frac{3a}{2}$$

$$e = \frac{c}{a} = \frac{1}{2} \Rightarrow c = \frac{a}{2}$$

$$r = R + h = R + \frac{R}{2} = \frac{3R}{2}$$

If $r_{min} = \frac{3R}{2} \Rightarrow a = 3R$

and if $r_{max} = \frac{3R}{2} \Rightarrow a = R$

If $a = 3R$, then, $C = \frac{a}{2} = \frac{3R}{2}$

$$r_{min} = (a - c) = \frac{3R}{2} = r_1$$

and $r_{max} = (a + c) = \frac{9R}{2} = r_2$

$L = \text{constant}$

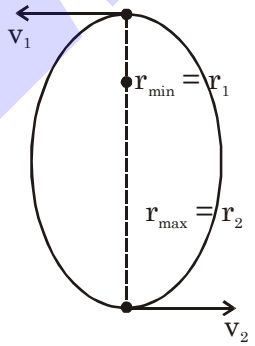
$mv_1r_1 = mv_2r_2$

$$v_1\left(\frac{3R}{2}\right) = v_2\left(\frac{9R}{2}\right)$$

$v_1 = 3v_2$

$TE = \text{conserved}$

$$\frac{1}{2}mv_1^2 - \frac{GmM}{r_1} = \frac{1}{2}mv_2^2 - \frac{GmM}{r_2}$$



$$\frac{v_1^2}{2} - \frac{GM}{\left(\frac{3R}{2}\right)} = \frac{\left(\frac{v_1}{3}\right)^2}{2} - \frac{GM}{\left(\frac{9R}{2}\right)}$$

$$\frac{v_1^2}{2}\left(1 - \frac{1}{9}\right) = \frac{GM}{R}\left(\frac{2}{3} - \frac{2}{9}\right)$$

$$v_1^2\left(\frac{4}{9}\right) = \frac{GM}{R}\left(\frac{4}{9}\right)$$

$$v_1^2 = \frac{GM}{R}$$

$$TE = \frac{1}{2}mv_1^2 - \frac{GMm}{r_1}$$

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$$= \frac{1}{2}m\frac{GM}{R} - \frac{GMm}{\left(\frac{3R}{2}\right)}$$

$$\Rightarrow \underset{(\text{finally})}{TE} = \frac{GmM}{R}\left(\frac{1}{2} - \frac{2}{3}\right) = \left[-\frac{GmM}{R}\left(\frac{1}{6}\right)\right]$$

So, change in TE required $= \frac{GmM}{6R}$

Similarly solving for (a = R)
then we will get

$$TE_{final} = -\frac{GmM}{2R}$$

So, change in $TE = \frac{GmM}{2R}$

Alternate Method
For perihilion

$$a(1-e) = R + h = \frac{3R}{2}$$

$\Rightarrow a = 3R$

Also, for elliptical path

$$T.E. = -\frac{GMm}{2a} = \frac{-GMm}{6R}$$

$\Rightarrow$ change in $TE = \frac{GMm}{6R}$

For aphelion :

$$a(1+e) = R + h = \frac{3R}{2}$$

$\Rightarrow a = R$

$$\& \text{ T.E.} = -\frac{GMm}{2a} = -\frac{GMm}{2R}$$

$\Rightarrow$ change in $TE = \frac{GMm}{2R}$

Linked Comprehension Type (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

16. Ans. (B)

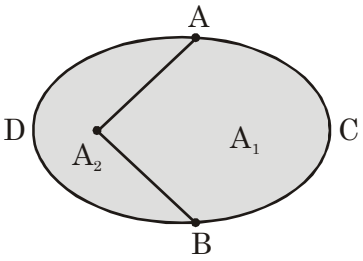
Sol. According to Keplar's Law

$$\frac{dA}{dt} = \text{constant}$$

$$\Rightarrow A \propto t \Rightarrow \frac{A_1}{A_2} = \frac{t_1}{t_2}$$

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$A_1 > A_2$

$\Rightarrow t_1 > t_2$

17. Ans. (C)

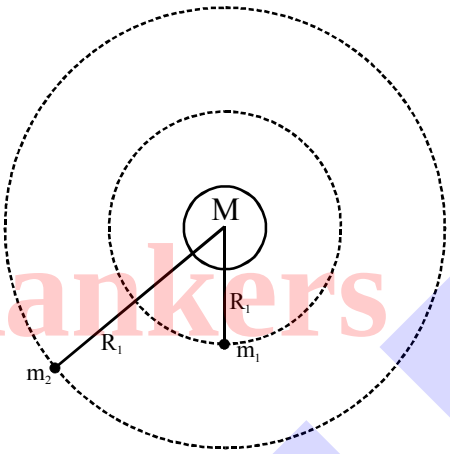
Sol. For a bounded system, TE < 0

$\Rightarrow |U| > |KE|$

Matching List Type (4 × 4) **1Q.[3 M (–1)]**

18. Ans. (B)

Sol.



$$\left.\begin{aligned}\frac{m_1}{m_2} &= 2\\ \frac{R_1}{R_2} &= \frac{1}{4}\end{aligned}\right\} \text{given}$$

$$\frac{GMm_1}{R_1^2} = \frac{m_1 v_1^2}{R_1}$$

$$v_1^2 = \frac{GM}{R_1}, \quad v_2^2 = \frac{GM}{R_2}$$

$$\frac{v_1^2}{v_2^2} = \frac{R_2}{R_1} = 4$$

(P) $\frac{v_1}{v_2} = 2$

(Q) $L = mvR$

$$\frac{L_1}{L_2} = \frac{m_1 v_1 R_1}{m_2 v_2 R_2} = 2 \times 2 \times \frac{1}{4} = 1$$

(R) $K = \frac{1}{2}mv^2$

$$\frac{K_1}{K_2} = \frac{m_1 v_1^2}{m_2 v_2^2} = 2 \times (2)^2 = 8$$

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(S) $T = 2\pi R/V$

$$\frac{T_1}{T_2} = \frac{R_1}{v_1} \times \frac{v_2}{R_2} = \frac{R_1}{R_2} \times \frac{v_2}{v_1} = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$$

SECTION-III

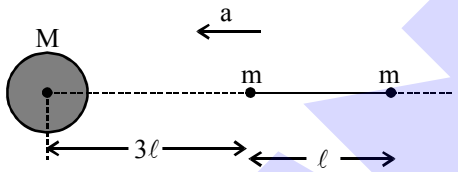
Numerical Grid Type (Ranging from 0 to 9) **2 Q. [4 M (0)]**

19. Ans. 6

Sol. $\frac{L_{total}}{L_B} = \frac{m_1 r_1^2}{m_2 r_2^2} + 1$

20. Ans. 7

Sol. Due to gravitational interaction connected masses have some acceleration.
Let both small masses are moving with acceleration 'a' towards larger mass M



Force eq. for mass nearer to larger mass

$$\frac{GMm}{(3\ell)^2} - \frac{Gm^2}{\ell^2} = ma \quad \text{... (i)}$$

Force eq. for mass away from larger mass

$$\frac{GMm}{(4\ell)^2} + \frac{Gm^2}{\ell^2} = ma \quad \text{... (ii)}$$

from equation (i) & (ii)

$$\frac{GM}{9\ell^2} - \frac{Gm}{\ell^2} = \frac{GM}{16\ell^2} + \frac{Gm}{\ell^2}$$

$$\Rightarrow \frac{M}{9} - \frac{M}{16} = m + m$$

$$\frac{7M}{144} = 2m$$

$$\Rightarrow m = \frac{7M}{288} = k\left(\frac{M}{288}\right)$$

$$\Rightarrow K = 7$$

SECTION-IV

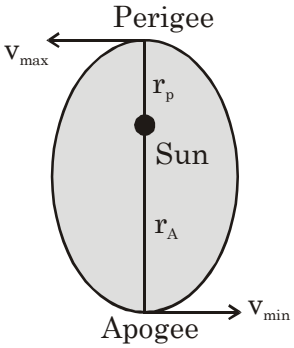
Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

21. Ans. (A)-R, (B)-Q, (C)-Q (D)-P

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Sol.



(A) $v \propto \frac{1}{r}$

(B) $r_p < r_A$

(C) $PE \propto -\frac{1}{r}$

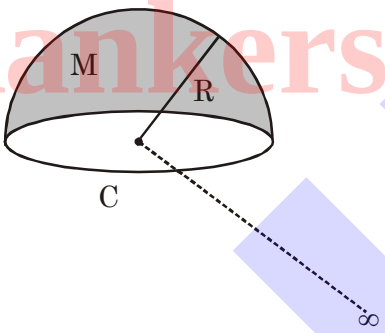
(D) L = constant

Subjective Type **5 Q. [4 M (0)]**

22. **Ans.** $A = -\frac{3}{2}\frac{GmM}{R}$

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Sol.



$W_g = \Delta U_g$

$= -m(\Delta V_g) = -m(V_{gf} - V_{gi}) = -m(V_\infty - V_C)$

$W_g = -m\left(0 - \left(\frac{-3GM}{2R}\right)\right) = -\frac{3GmM}{2R}$

23. **Ans.** (a) $T = \frac{2\pi d^{3/2}}{\sqrt{3Gm}}$, (b) 2, (c) 2

Sol. (a) The COM of the system divides the distance between the stars in the inverse ratio of thin masses. If d_1 & d_2 are the distances of M & m from the COM. (COM = Centre of mass)

$$d_1 = \frac{d}{M+m} \times m \ \& \ d_2 = \frac{d}{M+m} \times M$$

The stars will rotate in circles of radii d_1 & d_2 about their COM. The same force of attraction provides the necessary contripetal force for their circular motion.

$$\therefore \ \frac{GmM}{d^2} = M\omega_1^2 d_1 = m\omega_2^2 d_2$$

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or $\omega_1^2 = \frac{Gm}{d^2 d_1} = \frac{Gm}{d^2} \times \frac{M+m}{d \times m} = \frac{G(M+m)}{d^3}$

and $\omega_2^2 = \frac{G(M)}{d^2 d_2} = \frac{GM}{d^2} \times \left(\frac{M+m}{d \times M}\right) = \frac{G(M+m)}{d^3}$

$\Rightarrow \omega_1 = \omega_2 = \sqrt{\frac{G(M+m)}{d^3}}$

(b) From the fact that the moment of momentum is also angular momentum.

$$\frac{I_M}{I_m} = \frac{(Mv_1)d_1}{(Mv_2)d_2} = \frac{M}{m}\frac{d_1^2}{d_2^2} = \frac{M}{m} \times \frac{m^2}{M^2} = \frac{m}{M}$$

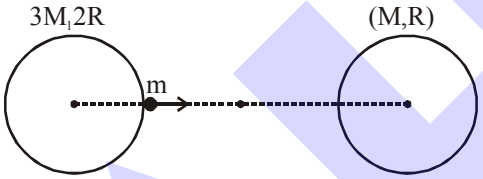
(c) $\frac{K_M}{K_m} = \frac{\frac{1}{2}Mv_1^2}{\frac{1}{2}Mv_2^2} = \frac{M}{m} \times \frac{\omega_1^2 d_1^2}{\omega_2^2 d_2^2} = \frac{M}{m}\frac{d_1^2}{d_2^2}$

$\Rightarrow \frac{K_M}{K_m} = \frac{M}{m} \times \frac{m^2}{M^2} = \frac{m}{M}$

24. **Ans.** $\left[2G(m_1+m_2)\left(\frac{2}{a}-\frac{1}{l}\right)\right]^{1/2}$

25. **Ans.** (a) 2R, (b) $3\sqrt{3}\left(\sqrt{3}-1\right)R$

Sol.



(a) speed will maximum at projection point

(b) To just reach the other planet small body must be able to reach the point where gravitation field due to both planet is zero. At this point speed will be minimum and equal to zero.

$$\frac{G \times 3M}{x^2} = \frac{GM}{(6R-x)^2} \Rightarrow \sqrt{3}(6R-x) = x$$

$$\Rightarrow \left(1+\sqrt{3}\right)x = 6\sqrt{3}R$$

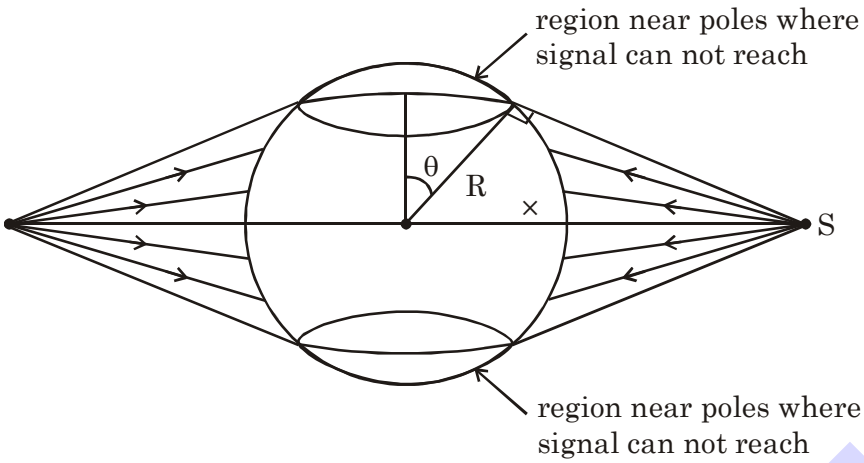
$$x = \frac{6\sqrt{3}\left(\sqrt{3}-1\right)R}{2}$$

26. **Ans.** $\left(1-\frac{\sqrt{x^2-R^2}}{x}\right)4\pi R^2$

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Sol.



$$\sin\theta = \frac{R}{x}$$

$$\cos\theta = \sqrt{1 - \frac{R^2}{x^2}}$$

Area required $= 2(2\pi R^2(1 - \cos\theta))$

$$= 4\pi R^2\left(1 - \frac{\sqrt{x^2 - R^2}}{x}\right)$$

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